

Introduction to LangChain

What is LangChain?

LangChain is an open-source framework for building LLM-powered applications. It provides standardized interfaces for models, prompts, chains, memory, agents, tools, document loaders, and vector stores.

Core Components

Models: Wrappers around LLMs (OpenAI, Anthropic, Google Gemini, Groq).

Prompts: Templates for structuring inputs to language models.

Chains: Sequences of calls combining models, prompts, and tools using LCEL.

Memory: State persistence across chain and agent calls.

Agents: Systems that use LLMs to decide which actions to take.

Document Loaders: Load data from PDFs, web pages, CSVs, directories, and more.

Vector Stores: Store and retrieve embeddings for semantic similarity search.

LangChain Expression Language (LCEL)

LCEL uses the pipe operator (`|`) to compose chains declaratively.

Example: `chain = prompt | llm | output_parser`

Benefits: easy streaming, parallel execution, async support, and tracing.

Why LangChain for RAG?

LangChain simplifies building Retrieval Augmented Generation pipelines. It connects document loaders, text splitters, embedding models, vector stores, and LLMs into a cohesive, reusable chain architecture.